Question 1 **2 marks** 101

Jack rolled a water bottle from one edge of his desk to the other. The desk measures 60 cm from edge to edge. The water bottle has a diameter of 7 cm. Determine the angle that the water bottle rotated, in degrees.

Question 2 2 marks 102

There are 15 dogs and 12 cats in an animal shelter. Determine the number of ways that three dogs and two cats can be selected if Scout, one of the dogs, must be selected.

Question 3 — 3 marks 103

Emily wants to save money to buy a car. She invests \$180 per month at an annual interest rate of 4.5%, compounded monthly.

Determine, algebraically, the number of monthly investments she will need to make to obtain at least \$15 000. Express the final answer as a whole number.

Use the formula: $FV = \dfrac{R\left[\left(1+i\right)^{n}-1\right]}{i}$

where $FV =$ the future value

$R =$ the investment amount each period

$i = \left[\dfrac{\text{the annual interest rate (as a decimal)}}{\text{the number of compounding periods per year}}\right]$

$n =$ the number of investments

Question 4 — 3 marks 104

Determine and simplify the 4th term in the binomial expansion of $\left(3x - \dfrac{2}{x^2}\right)^6$.

Question 5 3 marks 105

Solve, algebraically, over the interval $\left[0, 2\pi\right]$.

$$4\cos^2 x - 3\cos x - 1 = 0$$

Note: A calculator is not required for the remaining test questions.

Question 6 **1 mark** 106

State an equation for a rational function, $g(x)$, whose graph has a vertical asymptote at $x = 7$.

$g(x) =$ ________________________________

Question 7 **2 marks** 107

State an equation of a radical function, $f(x)$, with a domain of $x \leq 0$ and a range of $y \geq 1$.

$f(x) =$ ___________________________________

Question 8 — 3 marks 108

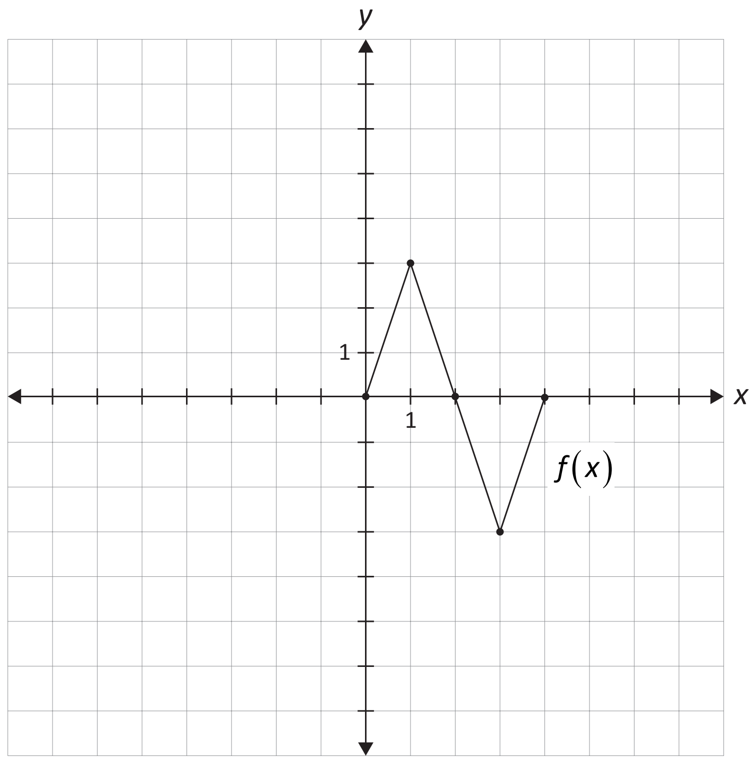
Prove the identity for all permissible values of x.

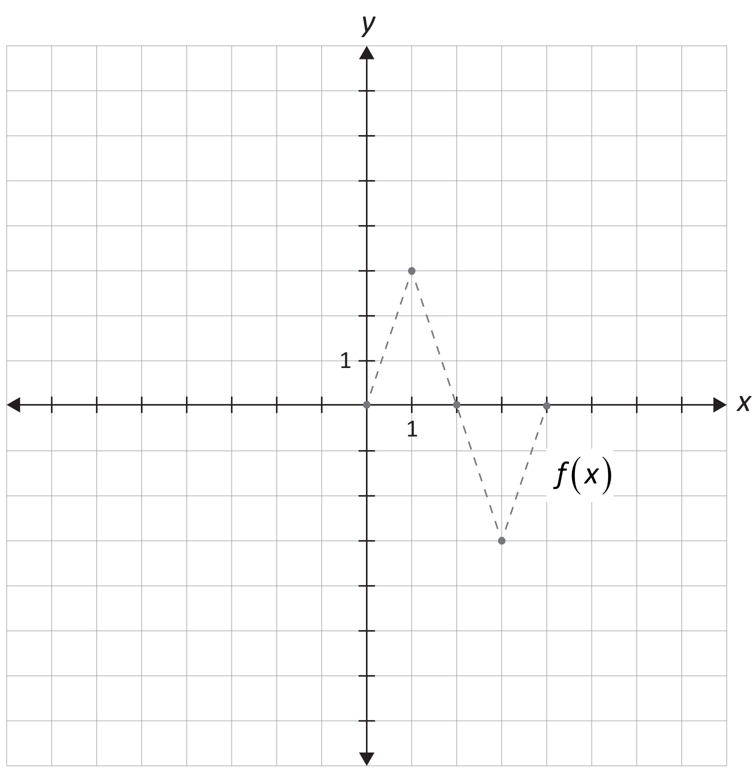
$$\frac{\cos x + \sin^2 x \sec x}{\sin x} = \sec x \csc x$$

Left-Hand Side	Right-Hand Side

Question 9 **3 marks** 109

Given the graph of, $y = f(x)$, sketch the graph of $y = 2|f(-x)|$.





The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 10   3 marks   110

Solve, algebraically.

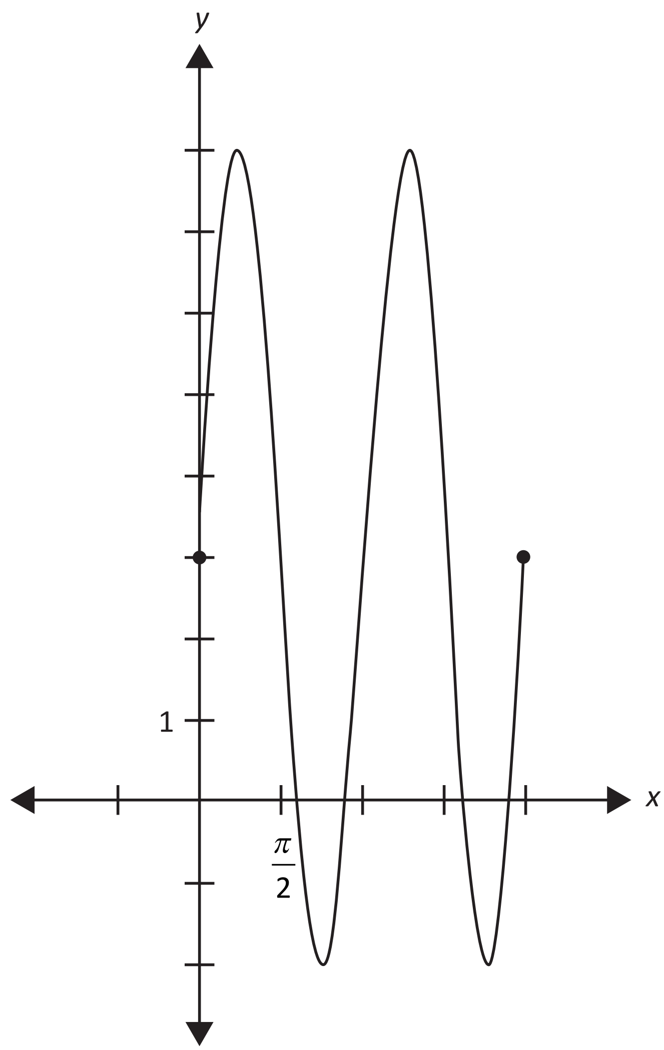
$$\log(x-1)+\log(x+2)=1$$

Question 11 **1 mark** 111

Justify that there are only two negative terms in the expansion of $\left(-2x+5y\right)^4$.

Question 12 **1 mark** 112

The graph of $y = 5\sin(2x) + 3$ below can be used to solve the equation $0 = 5\sin(2x) + 3$.
State how many solutions there are to the equation $0 = 5\sin(2x) + 3$ over the interval $[0, 2\pi]$.



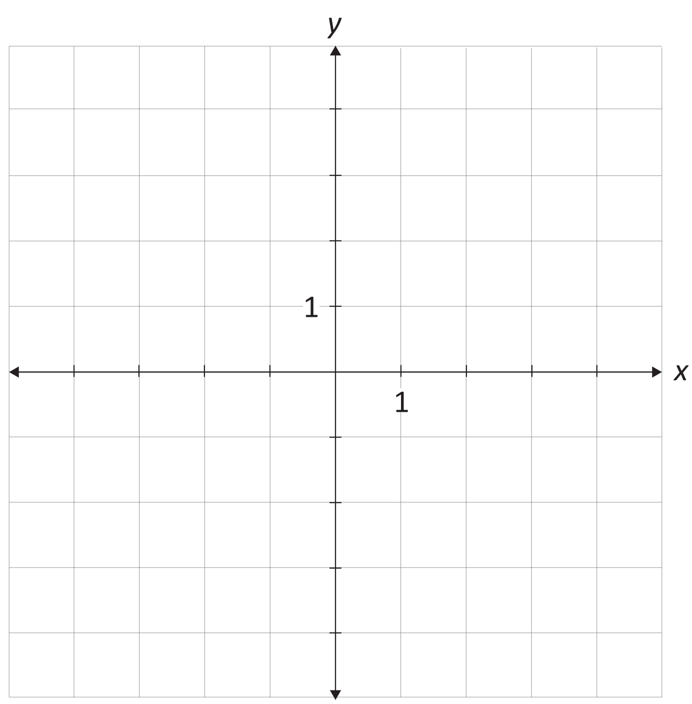
Question 13 **a) 1 mark b) 2 marks** 113 114

Given the functions, $f(x) = x^2 - 1$ and $g(x) = x + 1$,

a) state the equation of $h(x) = \dfrac{f(x)}{g(x)}$.

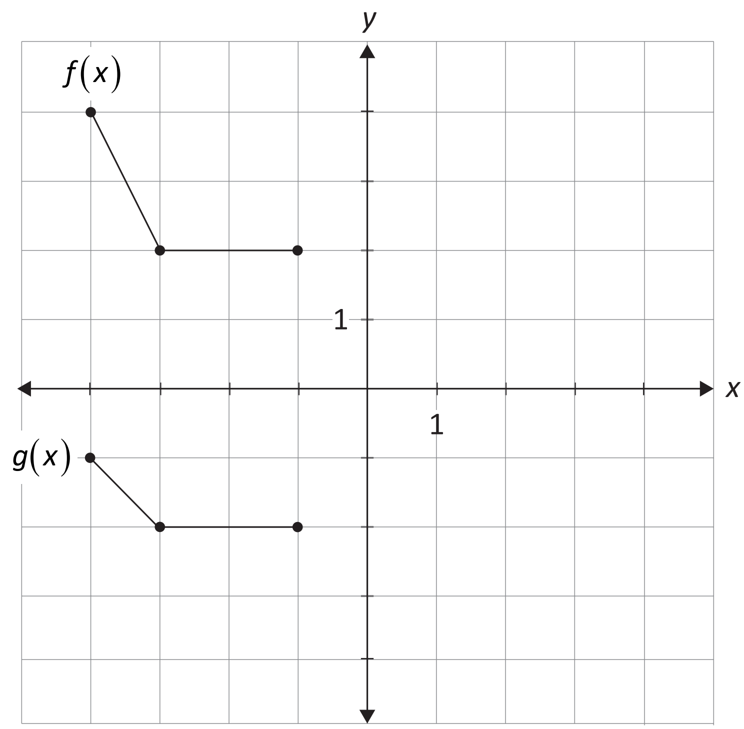
$h(x) =$ ____________________________________

b) sketch the graph of $h(x)$.



Question 14 — 2 marks

State an equation for $g(x)$, in terms of $f(x)$.



$g(x) =$ ____________________________________

Question 15 — 3 marks

116

Sketch the graph of a polynomial function, $p(x)$, with the following characteristics:

- degree 5
- leading coefficient of –1
- a zero at –3, with a multiplicity of 3
- a zero at 1, with a multiplicity of 2